GROKKING ACCELERATED: LAYER-WISE LEARNING RATES FOR TRANSFORMER GENERALIZATION

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ABSTRACT

This paper addresses the challenge of accelerating and enhancing the grokking phenomenon in Transformer models through layer-wise learning rates. Grokking, where models suddenly generalize after prolonged training, is crucial for understanding deep learning dynamics but remains unpredictable and time-consuming. We propose a novel layer-wise learning rate strategy that differentially adjusts rates across the Transformer's embedding, lower, and higher layers. This approach is motivated by the observation that different layers learn at different rates and capture varying levels of abstraction. Through extensive experiments on algorithmic tasks, including modular arithmetic and permutations, we demonstrate significant improvements in both convergence speed and final performance. Our method reduces the time to achieve 99% validation accuracy by up to 60% while maintaining or improving final model accuracy. Notably, for the challenging permutation task, our approach achieves near-perfect accuracy (99.95%) compared to the baseline's 3.59%. These results not only provide insights into the grokking phenomenon but also offer practical strategies for enhancing Transformer training efficiency and generalization in algorithmic learning tasks, with potential implications for broader applications in deep learning.

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1 INTRODUCTION

Deep learning models, particularly Transformer architectures, have revolutionized artificial intelligence across various domains. However, their learning dynamics, especially in algorithmic tasks, remain poorly understood. A fascinating phenomenon in this context is "grokking" Power et al. (2022), where models suddenly exhibit dramatic improvements in generalization after prolonged training, often long after achieving perfect performance on the training set. Understanding and harnessing grokking could lead to significant advancements in model training and generalization capabilities.

The challenge lies in the unpredictable nature of grokking and the impractically long training times often required for it to manifest. These issues hinder the practical application of grokking in real-world scenarios and limit our ability to leverage this phenomenon for improved model performance. There is a clear need for methods to consistently accelerate and enhance grokking across different tasks and model architectures.

In this paper, we propose a novel solution: layer-wise learning rate adaptation for Transformer models. Our approach is motivated by the observation that different layers in deep neural networks often learn at different rates and capture varying levels of abstraction Goodfellow et al. (2016). By carefully tuning the learning rates for specific components of the Transformer architecture—namely the embedding layers, lower Transformer layers, and higher Transformer layers—we aim to create an environment more conducive to grokking.

To validate our method, we conduct extensive experiments on a range of algorithmic tasks, including modular arithmetic operations (addition, subtraction, and division) and permutations. We implement a Transformer model in PyTorch Paszke et al. (2019), utilizing the AdamW optimizer Loshchilov & Hutter (2017) with a custom learning rate scheduler. Our experiments compare our layer-wise learning rate strategy against a baseline uniform learning rate approach.

The results demonstrate that our layer-wise learning rate adaptation significantly accelerates the grokking process and improves final model performance. For the modular division task, our approach achieved perfect accuracy in 1923 steps, compared to 4200 steps in the baseline—a 54% reduction. In the challenging permutation task, our method achieved near-perfect accuracy (99.95%) compared to the baseline's 3.59%. Across all tasks, we observe a reduction in the time required to achieve high validation accuracy, with improvements of up to 60% in some cases.

Our key contributions are:

- A novel layer-wise learning rate strategy for Transformer models that accelerates grokking in algorithmic learning tasks.
- Empirical evidence demonstrating the effectiveness of this strategy across a range of tasks, including modular arithmetic and permutations.
- Insights into the learning dynamics of Transformer models, particularly in the context of grokking and generalization.
- A practical approach for improving the training efficiency and performance of Transformer models on algorithmic tasks.

These findings open up several avenues for future research. Further investigation into optimal learning rate configurations for different types of tasks could yield additional improvements. Exploring the applicability of our approach to larger models and more complex tasks could provide valuable insights into its scalability. Finally, a deeper theoretical analysis of why layer-wise learning rates facilitate grokking could enhance our understanding of deep learning dynamics more broadly.

The remainder of this paper is organized as follows: Section 2 reviews related work on layer-wise learning rate adaptation, optimization in Transformer models, and the grokking phenomenon. Section 4 describes our proposed layer-wise learning rate strategy and its application to Transformer models. Section **??** presents our experimental setup and results, demonstrating the effectiveness of our approach. Finally, Section 7 concludes the paper and discusses potential future research directions.

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2 Related Work

Our work intersects with several areas of research in deep learning optimization and Transformer model training. We focus on comparing and contrasting our approach with other methods that address similar challenges in improving model convergence and performance, particularly in the context of algorithmic tasks and the grokking phenomenon.

Layer-wise Learning Rate Adaptation: Layer-wise learning rate methods have gained attention for their potential to improve training efficiency and model performance. Ko et al. (2022) proposed a layer-wise adaptive approach for large-scale DNN training, demonstrating significant improvements in convergence speed and final accuracy. Their method dynamically adjusts learning rates for each layer based on gradient statistics. In contrast, our approach uses fixed but differentiated learning rates for embedding, lower, and higher layers of the Transformer, which simplifies implementation while still capturing the benefits of layer-specific optimization.

Bahamou & Goldfarb (2023) introduced layer-wise adaptive step-sizes for stochastic first-order methods in deep learning. Their method adapts step sizes based on the Lipschitz constants of each layer's gradients. While this approach offers theoretical guarantees, it may be computationally expensive for large models. Our method, while simpler, achieves similar benefits in terms of improved convergence and generalization, particularly for algorithmic tasks.

Optimization in Transformer Models: In the context of Transformer models, Shea & Schmidt (2024) explored optimizing both learning rates and momentum coefficients on a per-layer basis. Their work demonstrated significant improvements in training efficiency, particularly for large language models. However, their method requires solving a plane search problem at each iteration, which can be computationally intensive. Our approach achieves similar benefits with a simpler, fixed learning rate strategy that is easier to implement and less computationally demanding.

Hu et al. (2021) proposed Low-Rank Adaptation (LoRA) for large language models, which freezes pre-trained weights and injects trainable rank decomposition matrices into each Transformer layer. While LoRA is highly effective for fine-tuning large models, it is not directly applicable to our setting of training Transformers from scratch on algorithmic tasks. Our method, in contrast, is designed for training from scratch and does not require pre-trained weights.

Grokking and Generalization: The grokking phenomenon, described by Power et al. (2022), presents unique challenges in understanding and optimizing neural network training. While Power et al. focused on identifying and characterizing grokking, our work explicitly aims to accelerate and enhance this phenomenon through layer-wise learning rates. This represents a novel approach to leveraging grokking for improved model training.

Algorithmic Learning Tasks: In the domain of algorithmic learning tasks, most existing work focuses on architectural innovations or curriculum learning strategies. Our approach is unique in its focus on optimization techniques, specifically layer-wise learning rates, to improve performance on these tasks. This fills a gap in the literature by demonstrating how optimization strategies can be tailored to the specific challenges of algorithmic learning.

Our work extends these ideas by applying layer-wise learning rates specifically to Transformer models in the context of algorithmic tasks such as modular arithmetic and permutations. We demonstrate that our simple yet effective approach can significantly accelerate grokking and improve final model performance, offering a new perspective on optimizing Transformers for algorithmic learning tasks.

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3 Background

Transformer models Vaswani et al. (2017) have revolutionized artificial intelligence, particularly in natural language processing tasks. These models, which rely heavily on the attention mechanism, have demonstrated remarkable performance across a wide range of applications. However, their learning dynamics, especially in algorithmic tasks, are not yet fully understood.

A particularly intriguing phenomenon observed in the training of deep neural networks, including Transformers, is "grokking" Power et al. (2022). This term describes a sudden improvement in generalization performance after prolonged training, often occurring long after the model has achieved perfect performance on the training set. Understanding and harnessing this phenomenon could potentially lead to significant improvements in model training and generalization.

Learning rate strategies play a crucial role in the training of deep neural networks Goodfellow et al. (2016). Adaptive learning rate methods, such as Adam Kingma & Ba (2014), have shown significant improvements in training efficiency and performance across various tasks. Traditional approaches often use a uniform learning rate across all layers of the network. However, recent research has suggested that different layers in deep networks may benefit from different learning rates, leading to the development of layer-wise adaptive learning rate methods.

Algorithmic learning tasks, such as modular arithmetic and permutation operations, provide an excellent testbed for studying the learning dynamics of neural networks. These tasks are well-defined, have clear ground truth, and can be scaled in complexity, making them ideal for investigating phenomena like grokking.

3.1 Problem Setting

In this work, we consider a Transformer model f_θ with parameters θ, trained on a dataset $D = \{(x_i, y_i)\}_{i=1}^{N}$, where x_i represents an input sequence and y_i the corresponding target output. The model is trained to minimize a loss function $L(f_\theta(x_i), y_i)$, typically cross-entropy for classification tasks.

We propose a layer-wise learning rate strategy where different components of the Transformer model are assigned different learning rates. Specifically, we define three groups of parameters:

- θ_e: parameters of the embedding layers
- θ_l: parameters of the lower Transformer layers